

TUTORIAL • OPEN ACCESS

A finite element method framework to model extracellular neural stimulation

To cite this article: Andreas Fellner *et al* 2022 *J. Neural Eng.* **19** 022001

View the [article online](#) for updates and enhancements.

You may also like

- [Spinal cord bioelectronic interfaces: opportunities in neural recording and clinical challenges](#)
Lei Jiang, Ben Woodington, Alejandro Carnicer-Lombarte et al.
- [Neural decoding of semantic concepts: a systematic literature review](#)
Milan Rybá and Ian Daly
- [Modeling extracellular stimulation of retinal ganglion cells: theoretical and practical aspects](#)
Kathleen E Kish, Scott F Lempka and James D Weiland



TUTORIAL

OPEN ACCESS

RECEIVED
27 October 2021

REVISED
9 March 2022

ACCEPTED FOR PUBLICATION
23 March 2022

PUBLISHED
7 April 2022

Original content from
this work may be used
under the terms of the
[Creative Commons
Attribution 4.0 licence](#).

Any further distribution
of this work must
maintain attribution to
the author(s) and the title
of the work, journal
citation and DOI.



A finite element method framework to model extracellular neural stimulation

Andreas Fellner^{1,*} , Amirreza Heshmat^{1,2} , Paul Werginz^{1,3,4} and Frank Rattay^{1,4}

¹ Institute for Analysis and Scientific Computing, Vienna University of Technology, Vienna, Austria

² Laboratory for Inner Ear Biology, Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria

³ Institute of Biomedical Electronics, Vienna University of Technology, Vienna, Austria

⁴ These authors jointly supervised this work.

* Author to whom any correspondence should be addressed.

E-mail: andreas.fellner@tuwien.ac.at

Keywords: finite element method, extracellular stimulation, neural stimulation, computer simulation, COMSOL

Supplementary material for this article is available [online](#)

Abstract

Objective. Increasing complexity in extracellular stimulation experiments and neural implant design also requires realistic computer simulations capable of modeling the neural activity of nerve cells under the influence of an electrical stimulus. Classical model approaches are often based on simplifications, are not able to correctly calculate the electric field generated by complex electrode designs, and do not consider electrical effects of the cell on its surrounding. A more accurate approach is the finite element method (FEM), which provides necessary techniques to solve the Poisson equation for complex geometries under consideration of electrical tissue properties. Especially in situations where neurons experience large and non-symmetric extracellular potential gradients, a FEM solution that implements the cell membrane model can improve the computer simulation results. To investigate the response of neurons in an electric field generated by complex electrode designs, a FEM framework for extracellular stimulation was developed in COMSOL. **Approach.** Methods to implement morphologically- and biophysically-detailed neurons including active Hodgkin-Huxley (HH) cell membrane dynamics as well as the stimulation setup are described in detail. Covered methods are (a) development of cell and electrode geometries including meshing strategies, (b) assignment of physics for the conducting spaces and the realization of active electrodes, (c) implementation of the HH model, and (d) coupling of the physics to get a fully described model. **Main results.** Several implementation examples are briefly presented: (a) a full FEM implementation of a HH model cell stimulated with a honeycomb electrode, (b) the electric field of a cochlear electrode placed inside the cochlea, and (c) a proof of concept implementation of a detailed double-cable cell membrane model for myelinated nerve fibers. **Significance.** The presented concepts and methods provide basic and advanced techniques to realize a full FEM framework for innovative studies of neural excitation in response to extracellular stimulation.

1. Introduction

Extracellular stimulation of nerve cells has a wide variety of applications in basic research, medical treatments, and the design of neuronal prostheses. The research of neural responses to electrical stimuli is focused on *in-vitro* and *in-vivo* experiments, which are often accompanied by computer simulations. The

computer simulations have a significant role in the interpretation of the experimental results, the prediction of the outcome of adapted experiments, and the translation of the experimental findings into possible technical strategies for treatments or neural implant design. With the increasing complexity of the experiments, also the requirements on the computer simulations increase.

The most common approach to simulate the neural response to electrical stimulation is combining a multi-compartment model with a model that computes the electric field generated by a stimulating electrode, either calculated by an analytical approximation, or by numerical approaches such as the finite element method (FEM). While such simulations are sufficiently accurate for many common problems, there are scientific questions beyond the capabilities of this model concept. Especially when the applied extracellular potential has large or non-symmetrical gradients, the simplifications in the geometry of the multi-compartment model may affect the accuracy of the results. Such strong extracellular potential gradients exist close to an electrode or in electrode designs which facilitate directed electric fields. Further, there are scientific questions that require a 3D representation of the extracellular space which also anticipates temporal changes in the electric field; examples are isotropic or anisotropic capacitive tissue properties as well as the influence of the spiking activity of a neuron on cells close by. Additionally, to investigate other effects coming along with extracellular stimulation (e.g. heat development, ionic diffusion, RC circuits behind the electrode) Multiphysics capabilities might be required. To address these and other scientific questions, a FEM framework capable of simulating the neural response to extracellular stimulation for realistic 3D cell and electrode geometries was implemented in COMSOL (www.comsol.com).

By inducing electrical stimuli in the vicinity of functioning neurons, their natural behavior of generating action potentials can be triggered. Examples for medical therapies based on extracellular stimulation are the treatment of Parkinson's disease (Benabid 2003), pain (Verrills *et al* 2016), movement after spinal cord injury (Minassian *et al* 2007), epilepsy (Fisher *et al* 2010), or depression (Mayberg *et al* 2005). While some applications are not very specific in which neurons are activated, other applications have much higher demands in targeting specific neurons at particular sites with the proper stimulus strength. To meet those requirements, a lot of research goes into the design of electrode systems (Naples *et al* 1988, Bucksot *et al* 2019, Flores *et al* 2019) and the investigation of cell-type specific activation (Meng *et al* 2018). Especially when extracellular stimulation is used to restore perception, as in cochlear implants (Bai *et al* 2019, Potrusil *et al* 2020) or retinal implants (Rattay and Resatz 2004, Werginz *et al* 2015), high spatial and temporal resolutions are necessary. In extensive studies, different stimulus setups and electrode designs are investigated with the ultimate aim to improve the realizable perception reached with neural implants. The specific objectives of these studies also affect the complexity of the assisting computer simulations. For many scientific and technical questions, simplified model approaches as commonly used cannot give satisfactory answers. Here, a FEM

framework is presented, which can deal with complex cell and stimulus geometries, and has the potential to answer complex biophysical questions in the context of neural stimulation.

A computer simulation for extracellular stimulation must apply a valid cell membrane model on a specific cell geometry under consideration of the stimulation setup (i.e. electrode geometry, stimulus parameters). Cell membrane models are mainly based on the formalism of the Hodgkin-Huxley (HH) model (Hodgkin and Huxley 1952), which describes the change in the cell membrane voltage as a function of time by a set of ordinary differential equations (ODEs). A HH-like model considers the kinetics of several voltage-sensitive ion channels and the capacitance of the cell membrane. A common technique to model the cell geometry is the multi-compartment model, where a cell is discretized into connected cylindrical or conical compartments with individual geometric and biophysical properties. This way, a closed electrical circuit is constructed, which can be evaluated by numerical integration methods (e.g. backward Euler method, Crank-Nicolson method) under consideration of Kirchhoff's laws. For this, free software solutions like NEURON (Hines and Carnevale 1997) are available and widely used. To model the stimulus, the electrical field as established by electrodes must be included in the electrical circuit of the multi-compartment model (i.e. calculate the extracellular potential at the outside of the cell during the stimulus). Due to the complexity of calculating the electric field, often simplifications like the point-source approach (i.e. a spherical electric field inside a homogeneous medium) or comparable approaches for disc electrodes are applied (McNeal 1976, Rattay 1990, 1999).

However, simplifications like the point source approach do not correctly reflect the electric field generated in the tissue, which is heavily influenced by the geometry of the electrodes and their carrier material, but also by the morphology and the respective electrical properties of the tissue (Joucla *et al* 2014). Furthermore, transmembrane currents due to neuronal activity and the presence of the neuronal structures themselves influence the potential distribution (Fellner *et al* 2019). Additionally, electrode impedances (Joucla *et al* 2014) and capacitive tissue properties (Grant and Lowery 2010) might influence the temporal development of the electric field during or after the stimulation. More accurate is the FEM approach, which can solve the Poisson equation of the electric field for complex geometries under consideration of distinct electrical properties and physical conditions.

Popular is the 'hybrid approach', where a FEM model for the electric field is connected to a multi-compartment cell membrane model (Joucla *et al* 2014, Potrusil *et al* 2020, Romeni *et al* 2020). The 'hybrid approach' allows to calculate the electric

field generated by electrodes in the FEM part, while the cell membrane model is applied in the simple and performant multi-compartment part. However, this concept also has its downsides. Two different software solutions are required (e.g. FEM software and in a second step NEURON), which need to be implemented and, in consequence, also maintained, either manually or by using a common interface (e.g. software link, shared data files). Due to two independent models, intermediate results of the multi-compartment model do not go into the FEM part and vice versa. It is common practice that only a static equilibrium of the electric field goes into the multi-compartment model and temporal aspects (e.g. capacitive tissue properties) affecting the electric field are not considered.

Another model strategy is the ‘whole FEM approach’ (Joucla *et al* 2014), where the active cell membrane model is also implemented in the FEM software. The monolithic design of the ‘whole FEM approach’ allows to model physical interactions between the neuron and its environment for every point in space and time. Depending on the FEM software, the simulation might consider additional physics, like electrode impedances or tissue permittivity. In the ‘whole FEM approach’, the cell geometry is fully realized in 3D and can be discretized to any degree of detail without having the geometric limitations as in a multi-compartment model (e.g. longitudinal vs. transversal current flow). However, the downsides of the ‘whole FEM approach’ are (a) high computational costs coming along when high spatial and temporal discretization is required, (b) more effort for model development, and (c) high costs of professional FEM software solutions.

A software to realize a ‘whole FEM approach’ must support methods to mathematically describe the cell membrane, such as COMSOL. Its Mathematics module allows an equation-based introduction of the HH cell membrane ODEs, its AC/DC module comes with all necessary interfaces to solve the Poisson equation of an electrical stimulus in complex domains. COMSOL provides a powerful geometry editor and different meshing methods but also supports importing geometries and meshes from other sources. Further, in COMSOL, physics can be coupled, which allows to include additional electrical constraints (e.g. frequency-dependent conductance, electrode-tissue interface, etc), and to model side effects such as heat development during the stimulation, which again affects the ion channel kinetics and thus the neuronal response.

Here, the concepts and methods to implement a FEM framework for extracellular neural stimulation in COMSOL are presented in detail. Particularly, the methods concentrate on (a) development of the geometry and meshing of electrodes and neurons, (b) setting up the extracellular stimulation, (c) implementation of a HH cell membrane model, and (d)

coupling of the physics. Further, techniques to reduce computational costs like smart meshing or intelligent time-stepping are outlined. Several example implementations are shown to underline the concepts. In the Results, a full FEM implementation of a HH cell membrane stimulated with a honeycomb electrode (Flores *et al* 2019) is found. Another example shows the field penetration of a cochlear implant in the cochlea. Due to the geometries of the electrodes in those examples, simplified model concepts are not capable of returning a realistic solution to such problems. Finally, a proof of concept implementation of the double-cable McIntyre-Richardson-Grill (MRG) model for myelinated fibers (McIntyre *et al* 2002) demonstrates that even complex cell membrane models are realizable with the presented FEM framework.

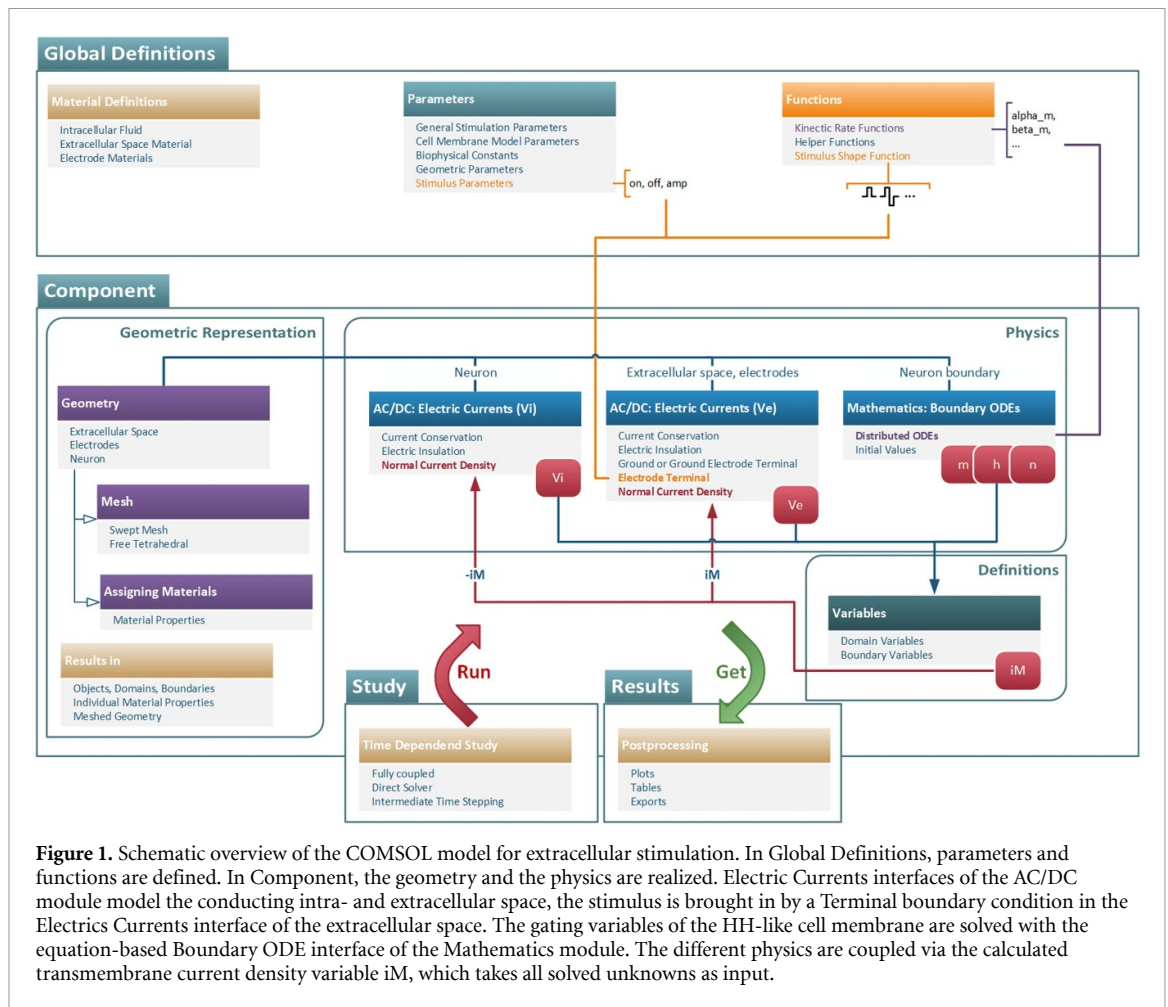
2. Methods

This section presents the basic concepts for implementing a 3D time-dependent ‘whole FEM’ computer simulation for extracellular stimulation in COMSOL. All methods were realized with COMSOL version 5.6 using the internal geometry editor and mesh builder. Applied physics are found in the AC/DC and the Mathematics modules.

The basic steps for FEM modeling are: (a) create or import a geometry, (b) mesh the geometry, (c) apply the physics including necessary parameters and boundary conditions, (d) solve the problem, and finally, (e) post-process the results. A COMSOL model is organized in a tree-view (figure 1) with sections for:

- **Global Definitions:** global parameters, functions, and geometry parts
- **Component:** variables and other definitions, the geometry, materials and their respective parameters, the applied physics with boundary conditions, and the mesh
- **Study:** solver settings and parameter sweeping
- **Results:** postprocessing like plots or export of data

A FEM model is always built up around the geometry of a problem. The basic geometric entities in a 3D COMSOL model are: points, edges, surfaces, and volumes; the corresponding physical entities are points, edges, boundaries, and domains. Selections allow to group or combine physical entities for easy access (e.g. a Selection that contains all the cell’s outer boundaries). Materials are used to store individual physical properties in the form of scalars, vectors, or matrices. Different shapes for 3D meshing are available (tetrahedrons, hexahedrons, triangular prisms, and pyramids). Automatic meshing algorithms can translate the whole geometry into a mesh at once (e.g. Free Tetrahedral mesh). However, as meshing highly influences the degree of freedom (DOF) and therefore the computational costs,



individual meshes for different domains eventually combined with swept meshes give better control. Predefined physics are organized in modules that again contain interfaces providing particular physical applications. Electric Currents (ec) interfaces of the AC/DC module are applied to model the conducting intra- and extracellular spaces; equation-based modeling as provided by the Mathematics module is used to implement a HH-like cell membrane. Different direct solvers which determine the actual runtime for a specific hardware system for fully coupled problems are available. It is also possible to configure a batch of simulations by applying parameter sweeps. Especially in the latter case, it is useful to export and store results after each run. As the actual solving might be quite time- and hardware-intensive, it is good practice to outsource the actual calculations to high-performance servers or clusters.

2.1. Global Definitions

Global parameters (e.g. biophysical membrane properties, geometric parameters, global model settings) are stored as Parameters by their name, unit, and value. COMSOL uses the International System of Units (SI); all inputs are automatically converted to the respective SI unit for further use. Parameters can be organized into logical groups; for each group, also

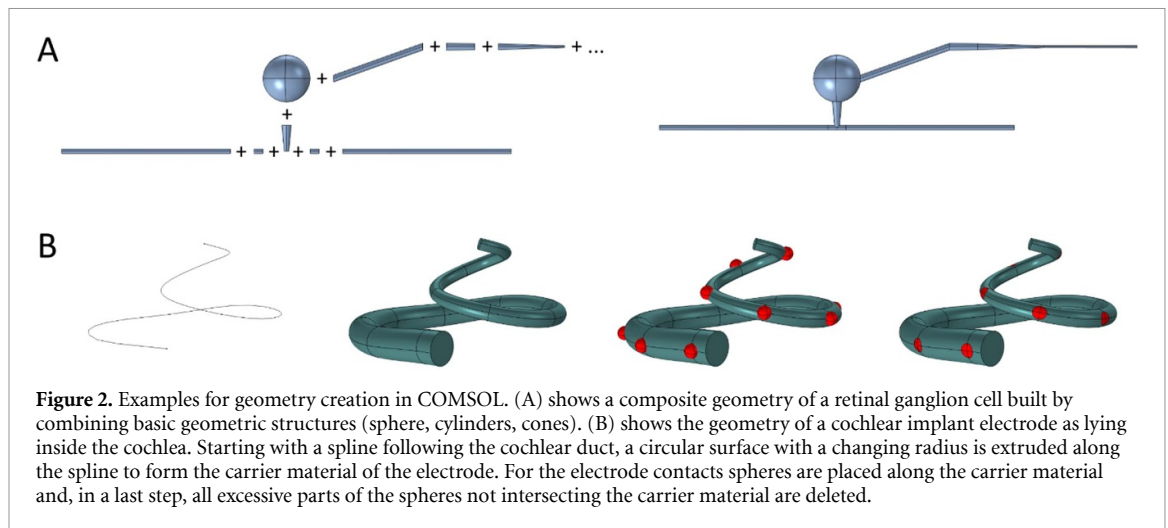
different sets of values (Parameter Cases) can be specified. Parameter Cases allow the implementation of parameterized variations (e.g. different geometries, variations in the biophysics, stimulus settings), which can either be loaded manually before solving or be swept by the solver in a batch run.

Further, in the Global Definitions also analytic or geometric functions are specified. Global functions as implemented in this model are:

- **Rate functions** (alphas and betas) of the HH-like cell membrane, which return the opening or closing probabilities of the ion channels based on the actual membrane voltage.
- **Trap function**, which is accessed by the rate functions to avoid divisions by zero.
- **Stimulus shape function** as a parameterized geometric function between 0 (stimulus off) and 1 (stimulus entirely on). Transition zones help to minimize the stiffness of the overall problem (e.g. rectangular stimulus shape with sinusoidal transition zones at the start and end).

2.2. Geometry/domains/materials

In COMSOL, a 3D geometry is represented by different domains having distinctive physical properties and physics. For extracellular stimulation, the



following domains are typically found: neuronal structure(s), electrodes, and the extracellular space. However, a geometric entity like an electrode or a neuron might consist of several adjacent domains for different parts (e.g. an electrode may consist of a metallic tip and the carrier material, a neuron of a soma and neurites).

A geometric entity can either be created by the internal geometry editor or imported from different sources (e.g. STL CAD files). Besides aligning basic geometric structures next to each other (e.g. figure 2(A): a model retinal ganglion cell built by cylinders, cones, and a sphere), the geometry editor also provides more sophisticated methods to create 3D structures (e.g. spline + extrude, see figure 2(B)). Different Boolean and transform operations support the creation of complex geometries; copy functions are available for repeating structures. Individual geometric entities can be stored as Geometry Parts and loaded for the final geometry, which is useful to realize geometric scalability.

By default, each geometric entity of the composite geometry will also become a physical domain. Uniting domains into logical coherent entities help to assign materials, physics, boundary conditions, and variables. Domains or boundaries can also be grouped into Selections, which bring more structure into the model and minimize errors in assignments. Examples for such Selections are: all domains that form the neuron or all outer boundaries of the neuron which stand for the cell membrane.

The level of details represented by the geometry has direct effects on the computational costs. Depending on the chosen resolution, the meshing algorithms try to preserve as many details of the geometry as possible by using small elements, which in consequence leads to a high DOF. Therefore, simplifications in the geometry might be conceived already when creating the geometry. Especially, long neuronal fibers or highly branched neurites can lead to a complexity that standard computers cannot handle

in an acceptable time. In this case, simplifications of the geometry, especially far away from the electrode, might help keep the DOF lower without having too much impact on the accuracy or the overall results.

2.3. Materials

Usually, to each domain a particular material is assigned. A material is a collection of material properties, which are required by the applied physics. A vast selection of common materials can be found in the internal material library. However, materials can also be created manually by adding them to the Global Definitions (e.g. materials for the intra- and extracellular space). When supported by the physics, material properties might not only be scalars but also tensors to specify anisotropic properties (e.g. anisotropic electrical conductivity in muscle fibers).

2.4. Meshing

The mesh determines the resolution of the geometry and, therefore, the DOF, an indicator of the computational costs of the model. By default, a tetrahedral mesh is applied by COMSOL for the whole geometry. The element shape and size and the discretization option (e.g. linear, quadratic) ultimately determine the number of nodes at which the unknowns of the physics are calculated. The DOF for a mesh consisting of quadratic discretized tetrahedrons (quadratic option not only considers nodes at all vertices, but also one additional node on each edge between two vertices) is approximately $\text{DOF} = 1.4 \cdot \# \text{elements} \cdot \# \text{unknowns}$. As each node is shared by several adjacent elements, this means that each additional tetrahedral element will cause approximately 1.4 additional nodes at which a solution for all unknowns (at each time step) needs to be calculated.

While the number of unknowns depend on the applied physics, the number of elements can be controlled in multiple ways. One method is the size definition, which can be defined for the whole mesh or only for parts (e.g. different meshing options of individual

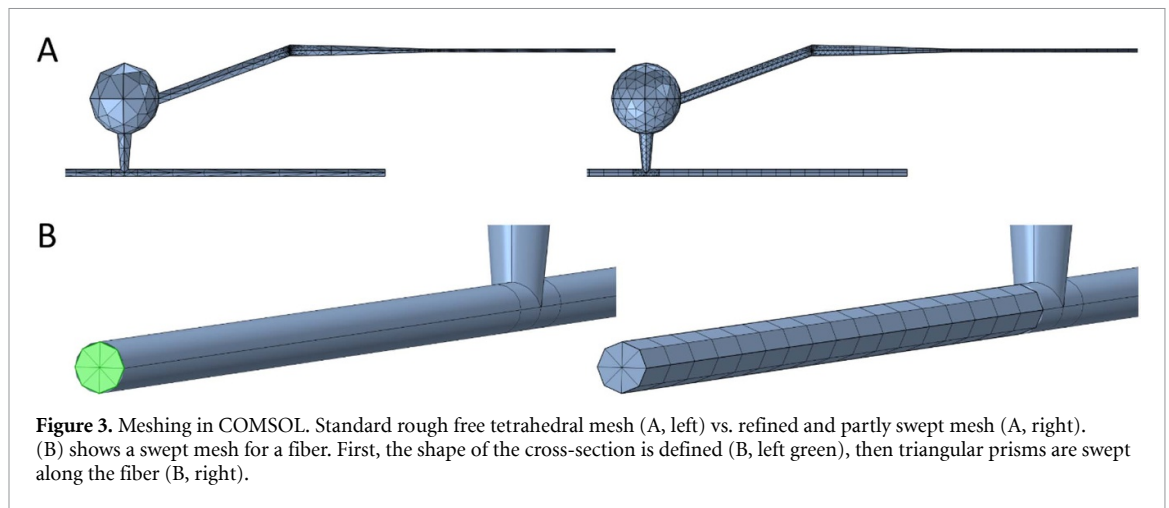


Figure 3. Meshing in COMSOL. Standard rough free tetrahedral mesh (A, left) vs. refined and partly swept mesh (A, right). (B) shows a swept mesh for a fiber. First, the shape of the cross-section is defined (B, left green), then triangular prisms are swept along the fiber (B, right).

domains). The size definition (minimum and maximum size, and properties for growth rate, curvature, and resolution of narrow regions) defines the accepted dimensions of a single element and the growth rate of elements in the volume. Elements are fitted from smaller elements in narrow regions to larger elements in wide regions. Another method is to choose between linear or quadratic discretization when supported by the respective physics; the linear option (no additional nodes at the elements' edges) reduces the DOF by a factor of approximately seven compared to the quadratic one.

Any reduction of elements is causing a loss in spatial resolution. Therefore, measures to optimize the mesh should be adapted with care and be related to the physics. For an extracellular stimulation of neuronal structures, the following points are to note:

- The strength of the electric field established by the electrodes decreases more or less with the square of the distance. Neuronal structures far away from an electrode experience a smaller gradient in the extracellular potential; this might justify a lower spatial resolution in the periphery without losing accuracy in the solution.
- A fiber shows an almost constant intracellular potential in the radial direction due to its low radial resistance. While a high resolution at the neuron's boundary in all directions is desired to anticipate possible extracellular potential gradients, no further refinement of the mesh towards the center of the fiber is required. The same is valid for large structures like a soma; due to its almost constant intracellular potential, the inside of the structure does not need a high spatial resolution.
- While neuronal structures are often simplified to cylinders or cones in modeling, in reality, they are far away from having that perfect shape. Especially for structures far away of the electrodes, a more or less round cross-section (i.e. octagonal or higher degree) might not improve the results further; instead, a hexagonal, quadratic, or

trigonal cross-section might be sufficient as long as the cross-sectional area is more or less preserved. By allowing a less round cross-section of fibers, the number of total elements can be reduced drastically.

- Besides the active sections with a high density of ion channels, a neuron usually also has passive or almost passive sections (e.g. passive dendrites, internodes in a myelinated fiber), which are not that sensitive to the extracellular stimulus. Those sections have an essential role in signal conduction but not in signal generation or amplification, which often justifies a rougher mesh in those sections.

The mesh creation can be performed separately for each domain utilizing different size and element shape definitions. It is possible to create a mesh where areas of special interest are meshed finer, while other regions that do not require a high resolution are meshed coarser (figure 3(A)). Especially for fibers, another technique of meshing is of particular interest, the swept mesh (figure 3(B)). For a swept mesh, the shape of the cross-section (e.g. octahedron out of triangles pointing to the center) is defined and consequently swept along the fiber in a way the cross-section stays preserved (e.g. by triangular prisms). This way, the number of elements along the fiber is fully configurable, which not only gives a high control over the mesh of the fiber but also over the mesh of the adjacent domains.

2.5. Electric field of the stimulus/extracellular space

The Electric Currents (ec) interface of the AC/DC module is used to solve the Poisson equation for the extracellular potential (V_e). The relevant biophysical properties are the electrical conductivity σ and the relative permittivity ϵ_r of the extracellular space. The current conservation ($\mathbf{J} = \sigma \mathbf{E}$) inside the medium also considers possible dielectric properties ($\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}$). The extracellular potential V_e is initialized to be zero ($V_e = 0$).

Both the electrical conductivity σ and the relative permittivity ϵ_r can be defined as scalars for isotropic tissues or tensors in an anisotropic medium. Anisotropy in conductivity or permittivity was observed in many types of tissue (Tuch *et al* 2001); for replicating the results of *in-vitro* experiments, usually, isotropic solutions fill the extracellular space (e.g. Ames' medium, artificial cerebrospinal fluid).

An active stimulus is modeled by a current or voltage Terminal boundary condition at the electrode tip domain. To include the stimulus properties (timing, shape, strength), a geometric shape function between 0 and 1 is multiplied with the stimulus strength (in Ampere or Volt) and set as value in the Terminal boundary condition. Defined transition zones (i.e. slightly ramping of the stimulus) of the shape function can be used to reduce the stiffness of the problem. An electrode can act as an active electrode or as ground. If there is no geometric representation of the ground electrode (e.g. ground is far away from any active electrode), the outer boundary of the extracellular space can act as a ground, which can also be defined to be in infinity. A concept where the ground is assumed in the infinity is closer to existing analytical approximations like the point source approach.

As displacement currents are considered by the HH-like cell membrane, the neuron itself must not be included in the extracellular space's Electric Currents (ec) interface. Instead, the boundaries between the neuron and the extracellular space are treated as isolating ($\mathbf{n} \cdot \mathbf{J} = 0$). The displacement currents are induced together with the ionic currents via the Normal Current Density boundary condition (see section 2.8).

2.6. Inner-axial currents/intracellular space

The conducting inside of the neuron is realized by another Electric Currents (ec) interface of the AC/DC module, which solves for the intracellular potential (V_i). The inside of a neuron is commonly assumed to be isotropic in conduction and permittivity. To establish the membrane voltage at rest (V_r), the intracellular potential is initialized by $V_i = V_r$. Entering or leaving transmembrane currents are implemented again with a Normal Current Density boundary condition.

2.7. Cell membrane model

The HH cell membrane model is implemented at the neuron's outer boundary and covers displacement currents due to the dielectric properties of the cell membrane and transmembrane ionic currents through ion channels. The displacement currents are determined by the capacitance of the cell membrane, the ionic currents depend on the opening/closing probabilities of the ion channels.

Variable definitions at the neuron's outer boundary are used to calculate the intermediate and

final results of the cell membrane model. The cell membrane voltage is determined by the intra- and extracellular potentials ($V_m = V_i - V_e$), which are directly accessible at the neuron's boundary. For simplicity, it makes sense to declare the actual membrane voltage as boundary variable: $V_m = V_i - V_e$.

The capacitive current density (i_c) is given by $i_c = c_m \dot{V}_m$, where c_m is the capacitance related to the area (nature constant: $c_m \approx 1 \mu\text{F cm}^{-2}$), and $\dot{V}_m$ is the time derivative of the membrane voltage. In COMSOL, this can be realized as a variable at boundary level by the term $i_c = c_m * d(V_m, t)$, where the function $d(\langle \text{expr} \rangle, t)$ returns the time derivative of the expression $\langle \text{expr} \rangle$.

The ionic current density $i_{\text{ion}} = \bar{g}_{\text{ion}} \prod x (V_m - E_{\text{ion}})$ of a particular ion channel, depends on the maximum conductance ($\bar{g}_{\text{ion}}$), the product of all associated gating variables (here denoted as x , in the original HH model: m and h for sodium ion channels, n for potassium), and the actual driving force in the form of $(V_m - E_{\text{ion}})$, where E_{ion} is the Nernst equilibrium potential. The Nernst potential E_{ion} accounts for the chemical diffusion force due to the imbalance of ion concentrations between the in- and outside of a cell.

The gating variables (original HH model: m , h , and n) of the ion channels in the form of $\dot{x} = \alpha_x (1 - x) - \beta_x x$ need to be calculated by solving the respective ODEs. Each gating variable x is based on the corresponding opening rate α_x and the closing rate β_x , which both are functions of the actual V_m and experimentally determined constants. These rates are defined as analytic functions in the Global Definitions. To avoid possible singularities due to divisions by zero, all denominators of the rate functions are checked for not being zero by a trap function. Each gating variable is treated as an additional unknown at the neuron's boundary by the Boundary ODEs and DAEs (bode) interface of the Mathematics module. The COMSOL Mathematics module provides several interfaces for equation-based modeling, which can also be applied to other types of ODEs commonly found in cell membrane models (e.g. ionic pumps).

The equation-based model concept requires defining the ODE as source term f with a respective initial value. Both the source term and the initial value are defined as expressions; therefore, other variables, parameters, or functions can act as input during solving. The unit of the source term is 1 s^{-1} , damping is set to 1, the mass coefficient to 0. In its most basic form, the source term of a state variable can be written as $\alpha_x(V_m) * (1 - x) - \beta_x(V_m) * x$, where $\alpha_x(V_m)$ and $\beta_x(V_m)$ are the corresponding opening and closing rate functions for the gating variable x . The source term can be extended by additional parameters, like temperature correction factors or other mathematical formulations, which influence the kinetics of an ion

channel. As an initial value, the steady-state value of a gating variable at resting voltage V_r is defined as: $\alpha_x(V_r)/(\alpha_x(V_r) + \beta_x(V_r))$.

The ionic current flux of each ion channel is calculated as a variable, which takes the actual gating states calculated by the Mathematics module into account (e.g. sodium ion channel of the original HH model in COMSOL: $i_{Na} = g_{Na} m^3 h^3 (V_m - E_{Na})$). The total transmembrane current density is given by $i_m = i_c + \sum i_{ion}$, which is then used to couple the physics of the intracellular potential V_i and the extracellular potential V_e (COMSOL: $iM = i_{lon} + iC$, where $i_{lon} = i_{Na} + iK + iL$).

2.8. Coupling of physics

To couple (i.e. electrically connect) the intracellular and extracellular space along the cell membrane, the total transmembrane current densities as calculated by the cell membrane model need to be considered by both physics. Notably, the currents flowing across the cell membrane are quite small, but while they have only a small effect on the usually large extracellular space, they have a strong impact on the small intracellular space. Nevertheless, for model consistency, any current flowing between the two domains should be treated accordingly on both sides of the neuron.

Therefore, each Electric Currents interface for V_i and V_e is extended by a Normal Current Density boundary conditions ($-\mathbf{n} \cdot \mathbf{J} = J_n$) which are used to induce the previously calculated transmembrane current density of the cell membrane model into both physics. Here it is essential to treat the sign of i_m accordingly. In HH-like models, positive currents are defined to flow out of the cell; the Normal Current Density boundary condition of COMSOL treats positive current densities as domain inward-directed. Thus, in the Electric Currents interface of the extracellular space, the respective Normal Current Density boundary condition is set up with the i_m as calculated by the cell membrane model ($V_e : J_n = i_m$), while in the interface of the intracellular space i_m must be treated with a negative sign ($V_i : J_n = -i_m$). In other words, a current entering the cell implies the same current leaving the extracellular space and vice versa.

2.9. Solver settings/time-stepping

COMSOL provides different types of studies; here, the time-dependent study is used. As there are no independent unknowns and all physics are coupled, the problem is treated as Fully coupled, which results in one large matrix of equations for all unknowns solved at once for each time step. Different direct solvers are available, which return the same results but, depending on the hardware and software of the machine COMSOL is running, might differ in actual time they need for solving the problem. In our environment, the

PARDISO-parallel sparse direct solver turned out to be the fastest solver.

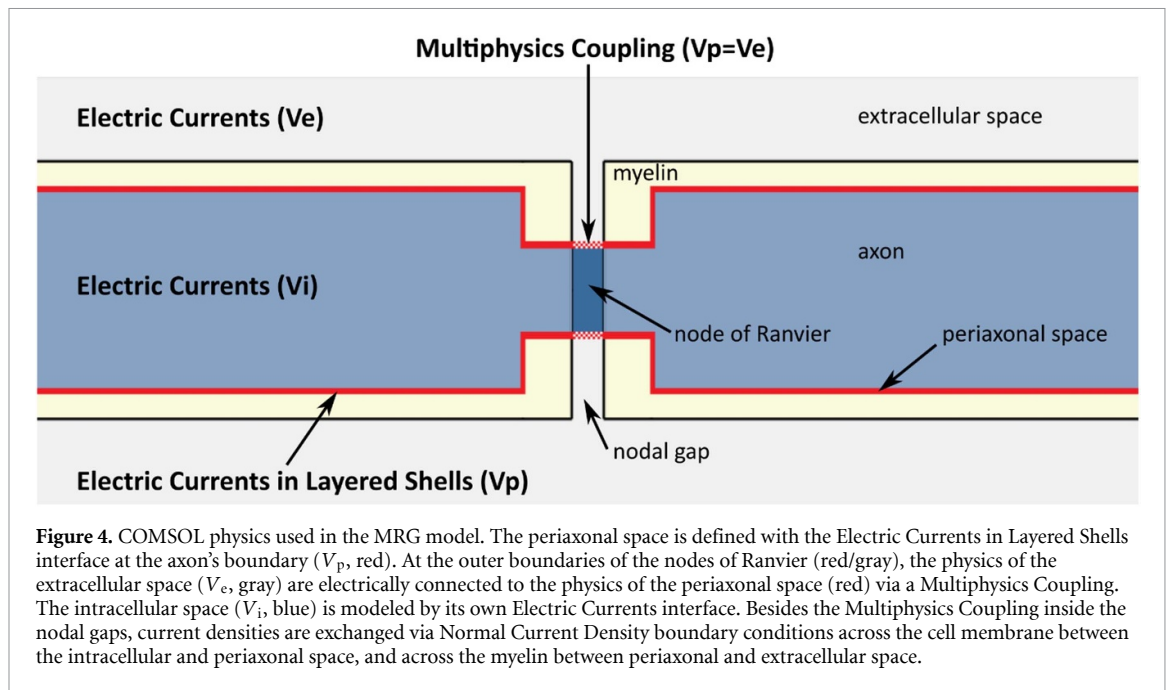
There are several settings according to tolerance and error handling, which depend on the individual requirements of the simulation. However, due to the stiffness of the problem caused by the only partly active stimulus, especially the time-stepping of the solver needs to be treated carefully. When using automatic time-stepping methods, the solver might miss the exact start or end of the stimulus. More control over the time-stepping is provided by the intermediate time-stepping method, a combination of automatic and manual time-stepping. With intermediate time-stepping, it is possible to enforce calculations at defined points in time, while in between those points, the solver can freely adjust its time steps. To control the temporal resolution of the simulation, the whole simulation period is divided into several time periods with individual temporal resolutions (COMSOL range(start time, step-size, end time) functions). Especially the sinusoidal transition zones of the stimulus shape function, which help to reduce the stiffness of the overall problem, are set up with a much higher temporal resolution than the rest of the simulation period. For long simulations or models that require some time to get in balance before the stimulation starts, also periods of a lower temporal resolution might be considered to save computational costs.

2.10. Advanced techniques for a double-cable model

The MRG model (McIntyre *et al* 2002) with its complex geometry provides several challenges when implementing in COMSOL. In-between the axon and the myelin sheets of the internodes, the extremely narrow periaxonal space is found. This space is electrically conducting and has openings to the nodal gaps. Because this space is several orders of magnitude thinner (2–4 nm) than any other geometric entity, a meshed 3D representation of it is not an option, and this situation needs a specific solution.

The Electric Currents in Layered Shells (ecis) interface of the AC/DC module provides the necessary physics to model a conducting thin layer at boundary level. This interface is continuously assigned along the whole axon (figure 4, also across the nodes of Ranvier) and models the electrical potential of the periaxonal space (V_p).

The resistance of the layer is determined by considering a specified height without having an actual volumetric representation. The Conductive Shell boundary condition with the Highly conductive layer option is used for current conservation, which implements the electrical conductivity and relative permittivity. Along the boundary of the internodal axons (i.e. axon's outer boundary without the nodes of Ranvier), Normal Current Density boundary conditions



are used to exchange currents between the intracellular and periaxonal space (i_m), and to exchange currents between the periaxonal and extracellular space across the myelin (i_{my}). Again, both current densities are induced with opposite signs in the respective physics for the intracellular, extracellular, and periaxonal potential ($V_i : J_n = -i_m$; $V_e : J_n = i_{my}$; $V_p : J_n = i_m - i_{my}$). The openings into the nodal gaps are realized with a COMSOL Multiphysics Coupling, which electrically connects the layer at the outer boundaries of the nodes of Ranvier with the extracellular space ($V_p = V_e$).

While in common models there is only one relevant boundary (i.e. the cell membrane), in the MRG model, there are two distinct boundaries separated by the myelin (i.e. the axon's boundary and the myelin's outer boundary). To provide access on variables only defined on the respective other boundary (e.g. access V_e at the axon's boundary from the myelin's outer boundary), General Extrusion operators are used. A General Extrusion operator maps variables across two boundaries by defining the geometric relation between source and destination (e.g. for a fiber lying in x-direction: source point (x, y, z) on the axon's boundary is mapped to the destination point $(x, r_{my}/r_a \cdot y, r_{my}/r_a \cdot z)$ at the myelin's outer boundary, where r_{my} is the radius of the fiber, r_a is the radius of the axon).

3. Results

Previously described model concepts have been used to implement several FEM models for different stimulation setups and cell types. Some of those simulations are briefly presented here to demonstrate

the potential of FEM modeling in computational neuroscience.

3.1. Standard HH model for a honeycomb electrode

As an exemplary implementation combining all mentioned model concepts, a standard HH cell membrane was implemented for a model neuron consisting of a soma and an axon only (figure 5). This model neuron lies outside of a single honeycomb electrode as used for electrode arrays in retinal implants. In the honeycomb electrode design, the active electrode is surrounded by isolating walls, where the ground electrode sits on top of these walls (Flores et al 2019). In contrast to bare electrode contacts as used in common electrode arrays, the honeycomb design produces a vertically directed electric field. The lower field penetration in the horizontal direction improves the targeting of specific cells and enables electrode arrays with a higher pixel resolution.

Due to the directed form of the electric field, a honeycomb electrode cannot be simulated with simplified model approaches or analytic approximations. Also, the geometry of a neuronal structure exposed to this kind of electric field cannot be adequately represented in a multi-compartment model, as it is impossible to align the compartments correctly (e.g. longitudinal vs. transversal current flow, no rotational symmetry to utilize).

In contrast, a FEM implementation can solve the Poisson equation for the complex electrode geometry and does not have a model direction, which results in a realistic polarization of the neuron due to the stimulus. The implementation features the passive (i.e. capacitance and leakage) as well as the active (i.e. ion channels) characteristics of the neuron.

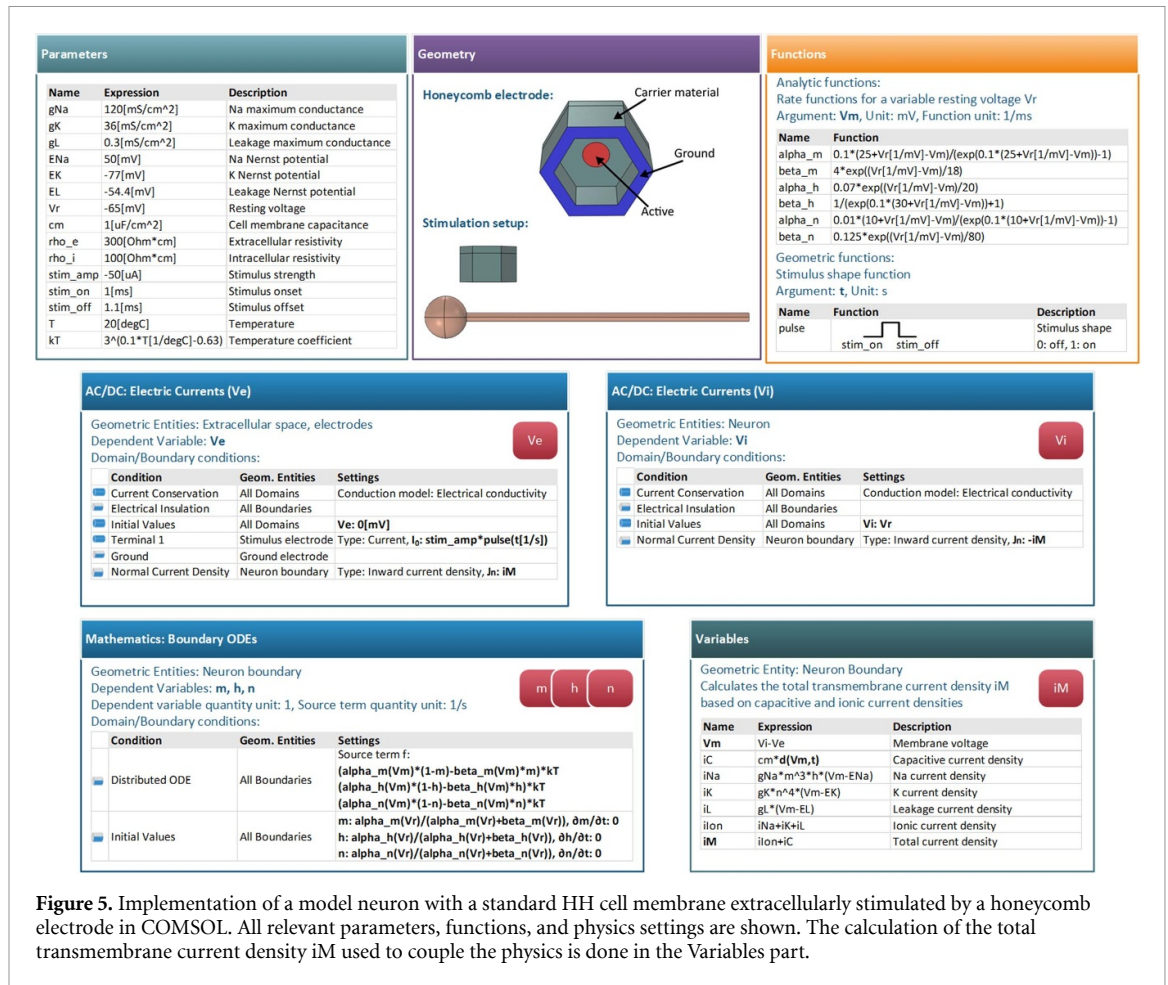


Figure 5. Implementation of a model neuron with a standard HH cell membrane extracellularly stimulated by a honeycomb electrode in COMSOL. All relevant parameters, functions, and physics settings are shown. The calculation of the total transmembrane current density iM used to couple the physics is done in the Variables part.

The final model can be used to investigate details on the activation of neurons stimulated with honeycomb electrodes; further, it can serve as a basis to couple it with additional physics.

The implementation is shown in detail in figure 5, which contains the relevant parameters and functions, the setup of the AC/DC physics for the intra- and extracellular space, the HH cell membrane implementation with the Mathematics module, and the calculation of the total transmembrane current density used to couple the different physics. The COMSOL file of this model is provided in the supplementary (available online at stacks.iop.org/JNE/19/022001/mmedia) and requires a COMSOL license to open. Besides the general implementation, this model also implements some advanced concepts in geometry creation, selection handling for model organization, a swept mesh for the axon, and the solver setting with intermediate time-stepping.

Figure 6 shows the influence of the honeycomb electrode on the cell's polarization during the stimulus. Notable are the non-symmetric polarizations of the soma during the stimulus, which could not be reproduced with a simplified approach for the electric field of the stimulus or the multi-compartment approach for the geometry. The stimulation shown in figure 6(A) causes a propagating action potential.

Figure 6(B) shows the same situation but with tripled stimulus amplitude, which results in a blocking condition. Due to the strong soma depolarization in regions close to the electrode, a sodium current reversal (outflowing sodium ions) further repolarizes the soma, known as the upper threshold phenomenon (Boinagrov *et al* 2012, Rattay 2014, Fellner *et al* 2019). Additionally, a strong hyperpolarization in the axon right to the electrode during the active stimulus prevents the propagation of an action potential in the axon, known as anodal surround block (Ranck 1975, Rattay and Aberham 1993).

A comparison of the honeycomb electrode example between the 'whole FEM approach' and the more classical 'hybrid approach' shows significant differences in most of the analyzed cases (figure 7). The reasons for those differences are the limitations and simplifications in the hybrid approach. These limitations of the multi-compartment approach can be addressed to some extent by arranging the compartments according to the electric field (see figure 7, different results for the rotating soma axis). However, as shown in figure 7, small changes to the somatic axis alignment can lead to substantially altered results; but even when the ideal axis orientation for the soma is found, there are still differences between the 'hybrid approach' and 'whole FEM' solutions.

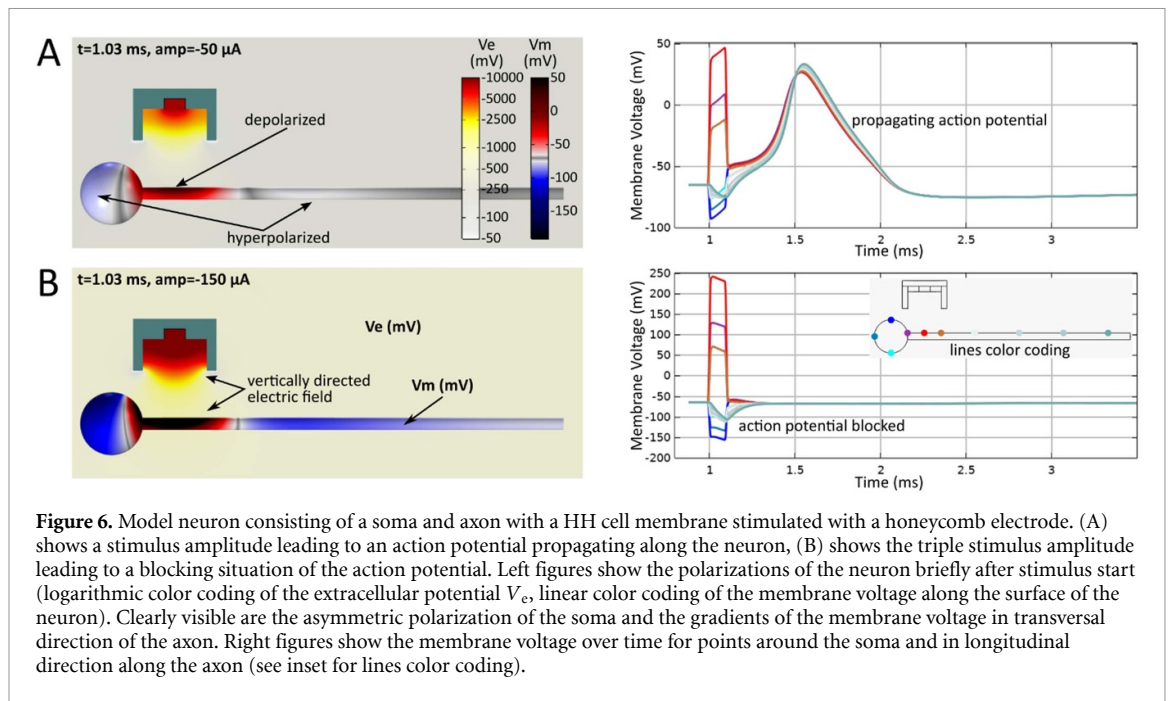


Figure 6. Model neuron consisting of a soma and axon with a HH cell membrane stimulated with a honeycomb electrode. (A) shows a stimulus amplitude leading to an action potential propagating along the neuron, (B) shows the triple stimulus amplitude leading to a blocking situation of the action potential. Left figures show the polarizations of the neuron briefly after stimulus start (logarithmic color coding of the extracellular potential V_e , linear color coding of the membrane voltage along the surface of the neuron). Clearly visible are the asymmetric polarization of the soma and the gradients of the membrane voltage in transversal direction of the axon. Right figures show the membrane voltage over time for points around the soma and in longitudinal direction along the axon (see inset for lines color coding).

The reasons for the partly substantial differences between the ‘hybrid approach’ and the ‘whole FEM approach’ are manifold. In the ‘hybrid approach’, the extracellular potentials passed to the multi-compartment model do not anticipate the presence of the cell itself (i.e. the potentials are calculated for the electrode in a homogeneous medium). Especially for cells close to an electrode, large cellular structures significantly influence the surrounding electric field, resulting in higher voltages at the cell membrane facing the electrode (Fellner *et al* 2019). Also, the multi-compartment model assumes a constant extracellular potential for every compartment, usually determined at the center of a compartment (Rattay 1999). This way of treating V_e ignores possible voltage gradients across a compartment. Further, depending on the diameter and the orientation of a compartment relative to the electric field, the V_e determined at the center of a compartment might strongly divert from the actual V_e acting on the outer surface; this is why the example of figure 7 is so sensitive to the somatic axis orientation. Lastly, the classical multi-compartment model has a clear direction of current flow given by the arrangement of the single compartments (longitudinal mode) and ignores any currents flowing in transversal direction which also have a significant contribution to the neural response (Meffin *et al* 2012).

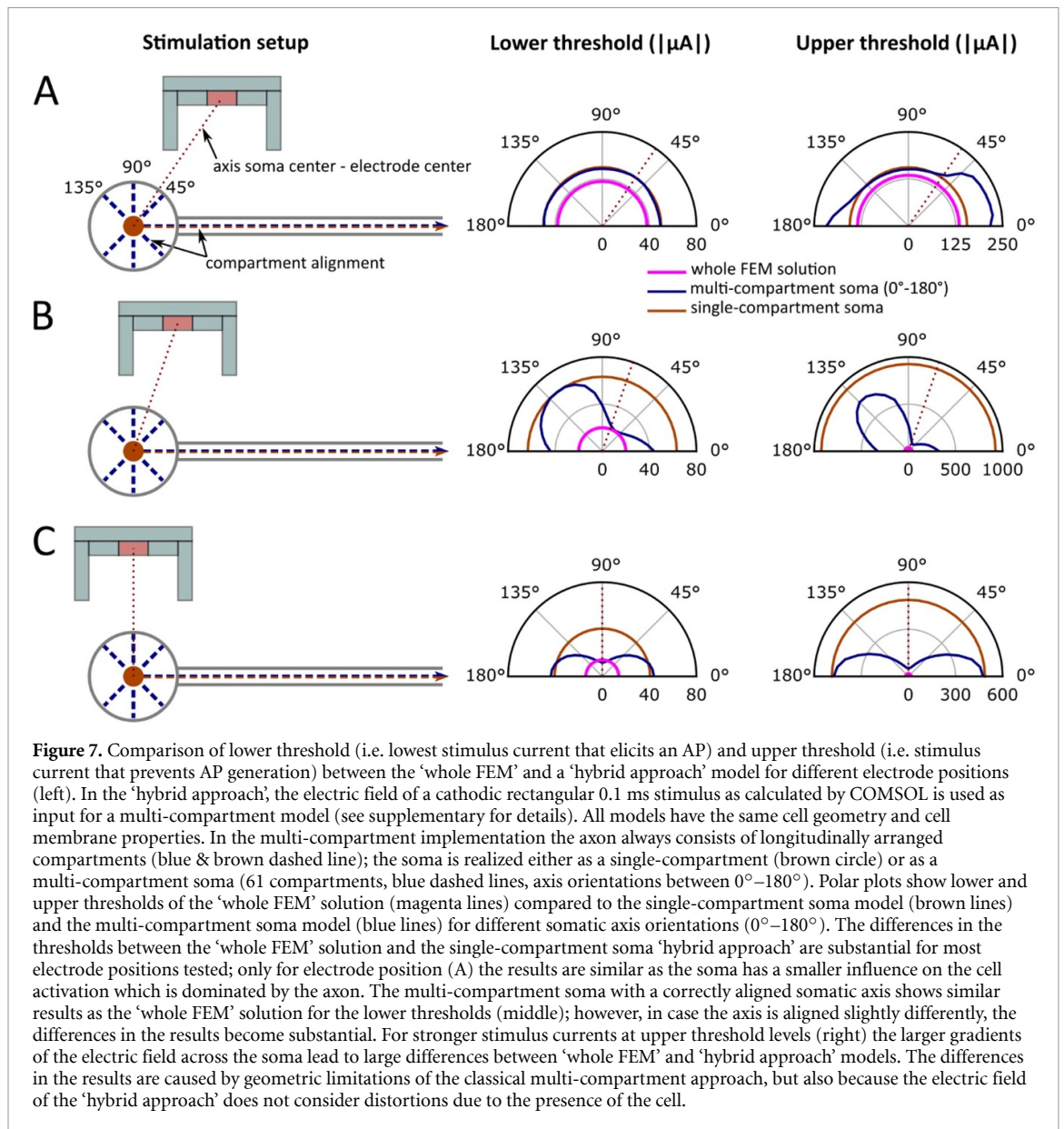
In contrast, the FEM provides the necessary numerical tools to solve physical interactions for complex geometric problems without having those limitations. The ‘whole FEM’ model accounts for the effects of the cell on the extracellular electric field caused by the cell’s presence and transmembrane currents. Due to the fine spatial discretization into small finite elements, voltage gradients in all directions are

considered. The extracellular potentials are taken directly from the cell’s boundary, and also dynamic changes in the V_e s are anticipated. Current flows are not limited in their direction, and distortions in the current flow due to junctions or bifurcations in the neural geometry are taken into account. Therefore, similar to other studies (e.g. Tahayori *et al* 2012), the ‘whole FEM’ model is seen as the reference solution in this example. It is possible to obtain closely matching results with the ‘hybrid approach’ by improving details in the modeling. However, without having a reference solution as given by the ‘whole FEM’ model, the actual error of the ‘hybrid approach’ is hard to assess.

3.2. Electric field of an electrode inside the cochlea

The next problem addressed with the FEM approach is the stimulation of the cochlea with a cochlear implant (in this example: MED-EL electrode array, www.medel.com). Inside the cochlea are several ducts filled with differently conducting fluids and the ducts are separated by poor conducting bone material; further, the isolating carrier material of the electrode array inserted into the cochlear influences the current flow inside the duct. These conditions demand a simulation approach such as FEM for a realistic simulation of the electric field within the cochlea.

In the COMSOL implementation, a 3D scan of the cochlea is imported from a SolidWorks file (Potrusil *et al* 2020). Different materials with corresponding electrical properties are assigned for the extracellular fluids and other tissue types found in- and outside the cochlea. The geometry of the flexible electrode is created (figure 2(B)) by considering both the electrode’s dimensions and the anatomical pathway of the cochlear duct. The electrode is realized by extruding



a circular surface with an inclining radius along a spline following the cochlear duct. For the extracellular stimulation, the imported cochlea and the electrode are brought together (figure 8(A)).

The electric field established by an active electrode (figure 8(B)) considers the respective geometries and electrical properties of the different cochlea parts, and the geometry and the isolating carrier material of the electrode itself. The resulting extracellular potential distribution is exported to a table (coordinates + electric potential) and further processed in a classical multi-compartment model implemented in MATLAB (Potrusil *et al* 2020). However, also a ‘whole FEM’ approach is thinkable, but since the size of the cochlea is several orders of magnitude larger than the diameters of the nerve fibers, a very fine mesh would be required. Instead of modeling the nerve fibers as 3D structures, a reduction of the nerve geometries to edges would drastically reduce

the complexity of the model (compare cuff electrode example from Dokos (2017)).

3.3. MRG model

The original HH cell membrane requires three additional ODEs for the gating variable (m , h , n) calculated with the Mathematics module, all other principles of the HH model are covered by the provided COMSOL physics. The same concept can be extended for more complicated models following the HH model formalisms, like the Fohlmeister retinal ganglion cell models (Fohlmeister and Miller 1997, Fohlmeister *et al* 2010, Fellner *et al* 2019) with its additional ion channels and an ion pump.

However, even more complicated models are available, which follow the HH model formalisms, but contain additional elements that are hard to implement in a FEM software. An example is the double-cable MRG model (McIntyre *et al* 2002) for

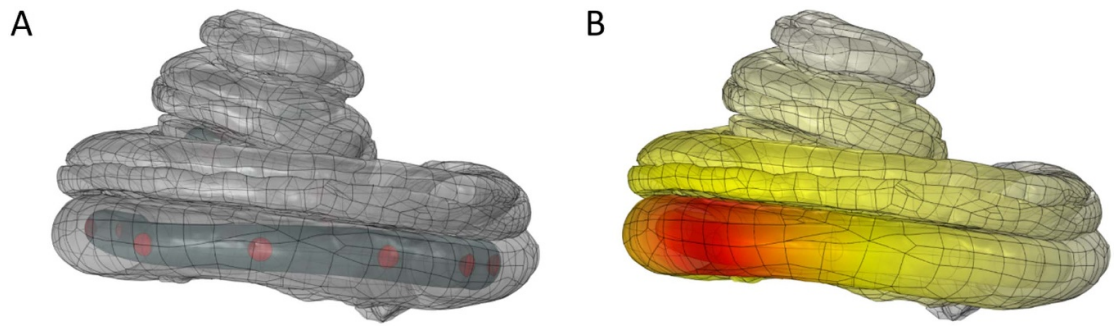


Figure 8. MED-EL electrode array placed inside the cochlea. (A) shows the geometry of a 3D scan of the cochlea with the electrode array inserted in the cochlear duct. The red circles are blank electrode contacts, the dark green volume is the carrier material of the electrode (compare figure 2(B)). (B) shows the extracellular potential (logarithmic color coding) created by an active electrode. The electric field penetrates differently along and across the anatomical parts of the cochlea. Notice the strong gradients in longitudinal direction of the cochlear duct compared to the weak gradients in transversal direction.

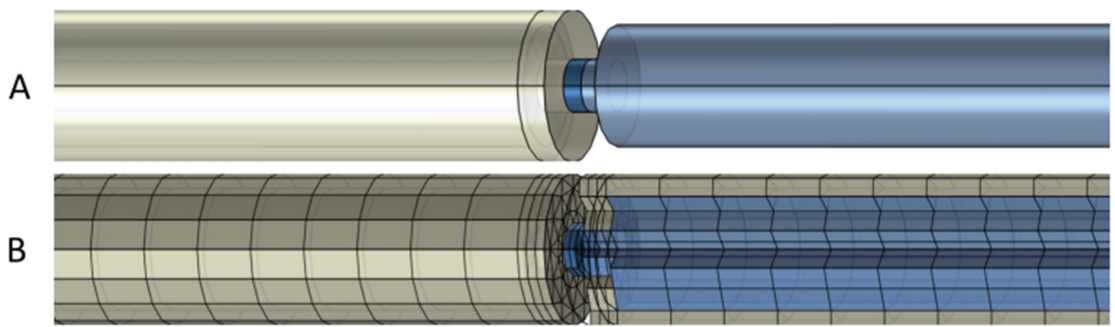


Figure 9. Geometry and swept mesh of the myelinated fiber. (A) shows the myelinated internode (left), the Node of Ranvier (center, dark blue), and the axon without a surrounding myelin sheet (right). (B) shows an example swept mesh for the myelinated fiber (left) and the cross-section of the swept meshed fiber (right).

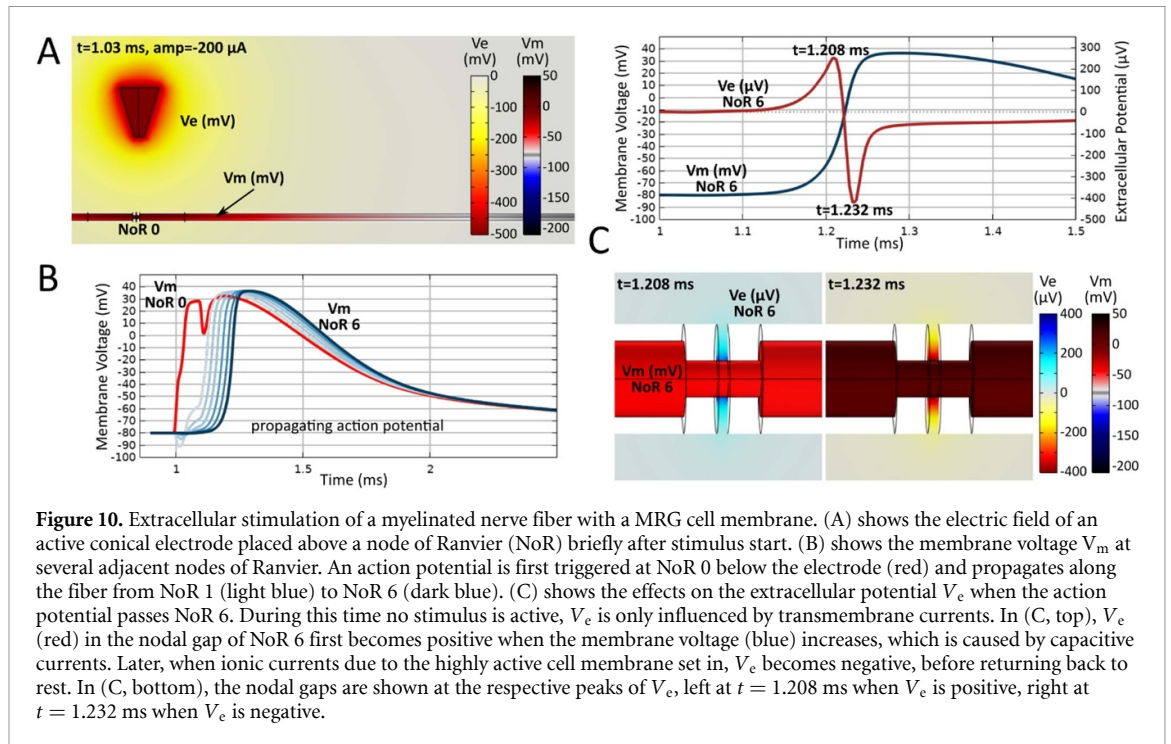
myelinated nerve fibers, which is commonly used for deep brain stimulation models. In the MRG model, the electrically conducting periaxonal space separates the axon from the surrounding myelin. The periaxonal space is several orders of magnitude thinner than the fiber, making a fully meshed 3D representation of this periaxonal space in a FEM model practically impossible. Another issue is the accessibility of all variables across different boundaries, which is necessary to implement all equations of the MRG cell membrane model. This proof of concept demonstrates how such a model with its additional constraints can still be realized in COMSOL by using advanced techniques to overcome those challenges.

The geometry of the ‘whole FEM’ MRG model does not include a geometric representation of the periaxonal space; only the fiber with its different sections and the surrounding myelin are modeled by cylinders (figure 9(A)). In the MRG model, the axon consists of several sections having different lengths and diameters. Thus, own geometric entities for a single node of Ranvier, a single internodal segment, and the myelin are constructed as Geometry Parts. Based on the wanted number of nodes of Ranvier, the whole fiber is assembled by copy operations, which replicates the single geometric entities and places

them accordingly. A swept mesh consisting of triangular prisms (figure 9(B)) is applied for the fiber. Depending on the section, different resolutions were chosen to minimize computations cost; the highly active nodes of Ranvier were higher resolved than the almost passive internodes.

To model the conducting periaxonal space (electrical potential V_p), the boundary of the whole axon (including the nodes of Ranvier) is declared as an electrically conducting layer by applying the Electric Currents in Layered Shells (ecis) interface of the AC/DC module to it. This layer is electrically connected with the extracellular space by a Multiphysics Coupling on the outer boundaries of the nodes of Ranvier. More details are described in the Methods (figure 4).

While this proof of concept shows that even such a complex model as the MRG model can be implemented in COMSOL (figure 10), it is actually quite a difficult task. To manage the complex geometry with all its different anatomical sections, an extremely high degree of organization in the model implementation is necessary; this was mainly established by defining Geometric Parts for the various geometric entities, by copy commands to assemble the fiber, and by the extensive use of boundary- and domain-Selections.



Parameter Cases have been proven beneficial to parameterize the fiber dimensions as specified in McIntyre *et al* (2002). The final model already quantifies the impact of the myelinated fiber on its environment (e.g. electric field in a nodal gap during the action potential, figure 10(C)) and is extendable for additional physics (e.g. RC circuit of a stimulator, tested but not shown).

4. Discussion

FEM modeling is a successful tool in many branches of biomedical engineering and basic research in the neurosciences. Most applications in functional electrical nerve stimulation include a boundary condition problem with respect to the body part of interest. Often, the problem is simplified in one or more aspects in modeling, like utilizing rotational symmetries (Boinagrov *et al* 2010, Meng *et al* 2018), reduction to a 2D problem (Reichel *et al* 1999), simplifications in the volume or nerve geometry (Kuhn *et al* 2010, Ladenbauer *et al* 2010), and, very commonly, analytic approximations for the electric field established by the electrodes (McNeal 1976, Fellner *et al* 2019, Werginz *et al* 2020).

While such simplifications are very useful in analyzing the general behavior of cells under the influence of an electrical stimulus, other problems require more accurate models. Examples are studies of electrode designs (Butson and McIntyre 2005, Datta *et al* 2008, Flores *et al* 2019), investigations of neuronal phenomena caused by the geometry of the electrode-tissue interface (Buitenweg *et al* 2002), or research of the electrical interaction between neural implants and

the tissue inside the body (Kase *et al* 2011, Potrusil *et al* 2020). All these examples are geometry-related boundary problems, for which FEM provides the necessary simulation tools.

A large variety of problems in neuroprosthetics have already been solved with the help of FEM, such as the impact of the electrode shape on the excitation of nearby passing nerve and muscle fibers (McIntyre and Grill 2001, Stickler *et al* 2009). A classical method exports FEM results of a stimulus into a multi-compartment model by its coordinates (Werginz *et al* 2015). Some investigations interested in the population size of excited nerve fibers when stimulated, calculate the second spatial derivative of the electric field with FEM to use the activating function (Rattay 1999) as a criterion (Bai *et al* 2019). Others simulate the field penetration of novel electrode designs in the tissue with the ultimate aim to increase achievable pixel resolution in electrode arrays (Flores *et al* 2019).

Typical for traditional investigations is to split the FEM analysis of stimulus fields from nerve fibers' excitation (Coburn 1985, Coburn and Sin 1985, Joucla *et al* 2014). Before comfortable FEM packages were available, the FEM was often restricted to a single element of constant size, and the simulation was performed with a custom-made program (Struijk *et al* 1991). Also, FEM software was usually limited to vendor-provided physics only. Nowadays, modern FEM software solutions like COMSOL can solve many interacting unknowns simultaneously, and concepts like equation-based modeling enable the manual implementation of additional physics, like the cell membrane.

The FEM allows to solve the Poisson equation of the electric field for complex geometries under consideration of distinct electrical properties and physical conditions. In combination with a cell membrane model, highly advanced simulations with interacting physics are realizable. The examples presented in the Results section show stimulation setups, which cannot be reproduced with other model concepts or analytical approximations. The ‘whole FEM’ honeycomb electrode example (figure 6) shows how a non-symmetric electric field generated by the specific electrode geometry acts on a model cell with HH membrane dynamics. This model was also benchmarked against the classical ‘hybrid approach’ (figure 7); the comparison shows that even with advanced concepts applied on the multi-compartment model, in some cases it is difficult to come close to the reference solution provided by the ‘whole FEM’ model. Another example of the cochlear implant demonstrates the field penetration of a stimulus inside a complex composite geometry with different electrical properties (figure 8). The MRG model example (figure 10) proves that the concepts and methods presented in this study can also be applied to very detailed cell membrane models. Additionally, the impact of the neuron’s own activities on its environment can be quantified (figure 10(C)), which is neglected in most modeling concepts.

The presented FEM framework is capable to simulate the neural response to extracellular stimulation for realistic 3D cell and electrode geometries in COMSOL. The concepts behind the framework are of general nature and not restricted to a particular setup. The cell membrane model for neural cell structures follows the same principles as in other model approaches and can be adapted to all kinds of HH-like cell membrane models; but, independent of the source, it takes the electrical potential of its environment into account. The same is valid for the extracellular space which is realized by standard COMSOL AC/DC physics with the only extension to account for transmembrane currents generated by the electrical activity of neurons.

As the interface between cell and surrounding was kept general by just exchanging current densities across the boundaries, the framework is open for expandability. For example, considering isotropic or anisotropic capacitive tissue properties is just a matter of setting the respective parameters in the COMSOL physics. Additional physics can be coupled via variables or via coupling interfaces available for selected physics in COMSOL. The physics coupling allows detailed investigations of other effects coming along with extracellular stimulation (e.g. heat development, ionic diffusions) or to implement other electrical phenomena of the electrode-tissue interface (e.g. Helmholtz layer, RC circuits at the electrode). Further, the same concepts as discussed here for a single neuron can be applied to multiple neurons, which allows

the investigation of cellular populations. Electrical interactions between neurons and also other effects like distortions in the extracellular electrical field are treated correctly and do not require additional efforts in programming.

However, this concept also has downsides that need to be addressed. One aspect is the high costs of the software. While commonly used software solutions for neural modeling such as NEURON are free of charge, the licensing costs of COMSOL are high. Also the computational costs that increase with spatial and temporal resolution are high. Intelligent meshing and time-stepping concepts can help to reduce the computational costs dramatically, but in comparison to other model concepts, the runtime of a FEM model is always longer. Taking the MRG model as an example, depending on the spatial and temporal dimensions and resolutions, the runtime for the ‘whole FEM’ MRG model, will be in the order of several hours. Solving the same model with a multi-compartment model takes only a couple of seconds. While such long runtimes are acceptable for smaller investigations, systematic investigations on complex problems usually have very high demands for the hardware, which again increases the overall costs. Finally, the development of sophisticated FEM models requires special skills and knowledge; in the context of neural simulation, the available literature is rare.

Taking the discussed advantages and disadvantages into account, the question remains when to use a FEM solution to simulate the response behavior of neurons. This question is difficult to answer in general, but some rules of thumb exist when the ‘whole FEM’ solutions is an option. First, situations where strong and non-uniform electrical gradients act on one or more cells which cannot be addressed adequately with a multi-compartment model. Second, studies on the electrical interactions between cells require a realistic 3D representation of the extracellular space which can treat current flow in tightly packed tissue. Third, investigations of other geometric-related phenomena occurring in the context of neural activity can be realized in COMSOL due to its Multiphysics simulation capabilities. And finally, as shown in figure 7, the ‘whole FEM’ solution might also act as a reference solution to set up a ‘hybrid approach’ model correctly.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

This research was supported by the Austrian Science Fund (FWF), Grant No. 29650. Further, we would like

to thank the COMSOL support team for their help and discussions.

ORCID iDs

Andreas Fellner  <https://orcid.org/0000-0003-3704-8764>

Amirreza Heshmat  <https://orcid.org/0000-0002-2052-152X>

Paul Werginz  <https://orcid.org/0000-0002-3441-3167>

Frank Rattay  <https://orcid.org/0000-0002-2819-8827>

References

- Bai S, Encke J, Obando-Leitón M, Weiß R, Schäfer F, Eberharter J, Böhnke F and Hemmert W 2019 Electrical stimulation in the human cochlea: a computational study based on high-resolution micro-CT scans *Front. Neurosci.* **13** 1312
- Benabid A L 2003 Deep brain stimulation for Parkinson's disease *Curr. Opin. Neurobiol.* **13** 696–706
- Boinagrov D, Loudin J and Palanker D 2010 Strength–duration relationship for extracellular neural stimulation: numerical and analytical models *J. Neurophysiol.* **104** 2236–48
- Boinagrov D, Pangratz-Fuehrer S, Suh B, Mathieson K, Naik N and Palanker D 2012 Upper threshold of extracellular neural stimulation *J. Neurophysiol.* **108** 3233–8
- Bucksot J E, Wells A J, Rahebi K C, Sivaji V, Romero-Ortega M, Kilgard M P, Rennaker R L and Hays S A 2019 Flat electrode contacts for vagus nerve stimulation *PLoS One* **14** e0215191
- Buitenweg J R, Rutten W L and Marani E 2002 Extracellular stimulation window explained by a geometry-based model of the neuron-electrode contact *IEEE Trans. Biomed. Eng.* **49** 1591–9
- Butson C R and McIntyre C C 2005 Role of electrode design on the volume of tissue activated during deep brain stimulation *J. Neural Eng.* **3** 1
- Coburn B 1985 A theoretical study of epidural electrical stimulation of the spinal cord—part II: effects on long myelinated fibers *IEEE Trans. Biomed. Eng.* **BME-32** 978–86
- Coburn B and Sin W K 1985 A theoretical study of epidural electrical stimulation of the spinal cord part I: finite element analysis of stimulus fields *IEEE Trans. Biomed. Eng.* **BME-32** 971–7
- Datta A, Elwassif M, Battaglia F and Bikson M 2008 Transcranial current stimulation focality using disc and ring electrode configurations: FEM analysis *J. Neural Eng.* **5** 163
- Dokos S 2017 *Modelling Organs, Tissues, Cells and Devices: Using MATLAB and COMSOL Multiphysics* (Berlin: Springer)
- Fellner A, Stiennon I and Rattay F 2019 Analysis of upper threshold mechanisms of spherical neurons during extracellular stimulation *J. Neurophysiol.* **121** 1315–28
- Fisher R et al 2010 Electrical stimulation of the anterior nucleus of thalamus for treatment of refractory epilepsy *Epilepsia* **51** 899–908
- Flores T, Huang T, Bhuckory M, Ho E, Chen Z, Dalal R, Galambos L, Kamins T, Mathieson K and Palanker D 2019 Honeycomb-shaped electro-neural interface enables cellular-scale pixels in subretinal prosthesis *Sci. Rep.* **9** 1–12
- Fohlmeister J F, Cohen E D and Newman E A 2010 Mechanisms and distribution of ion channels in retinal ganglion cells: using temperature as an independent variable *J. Neurophysiol.* **103** 1357–74
- Fohlmeister J F and Miller R F 1997 Impulse encoding mechanisms of ganglion cells in the tiger salamander retina *J. Neurophysiol.* **78** 1935–47
- Grant P F and Lowery M M 2010 Effect of dispersive conductivity and permittivity in volume conductor models of deep brain stimulation *IEEE Trans. Biomed. Eng.* **57** 2386–93
- Hines M L and Carnevale N T 1997 The NEURON simulation environment *Neural Comput.* **9** 1179–209
- Hodgkin A L and Huxley A F 1952 A quantitative description of membrane current and its application to conduction and excitation in nerve *J. Physiol.* **117** 500–44
- Joucla S, Glière A and Yvert B 2014 Current approaches to model extracellular electrical neural microstimulation *Front. Comput. Neurosci.* **8** 13
- Kuhn A, Keller T, Lawrence M and Morari M 2010 The influence of electrode size on selectivity and comfort in transcutaneous electrical stimulation of the forearm *IEEE Trans. Neural Syst. Rehabil. Eng.* **18** 255–62
- Ladenbauer J, Minassian K, Hofstoetter U S, Dimitrijevic M R and Rattay F 2010 Stimulation of the human lumbar spinal cord with implanted and surface electrodes: a computer simulation study *IEEE Trans. Neural Syst. Rehabil. Eng.* **18** 637–45
- Mayberg H S, Lozano A M, Voon V, McNeely H E, Seminowicz D, Hamani C, Schwab J M and Kennedy S H 2005 Deep brain stimulation for treatment-resistant depression *Neuron* **45** 651–60
- McIntyre C C and Grill W M 2001 Finite element analysis of the current-density and electric field generated by metal microelectrodes *Ann. Biomed. Eng.* **29** 227–35
- McIntyre C C, Richardson A G and Grill W M 2002 Modeling the excitability of mammalian nerve fibers: influence of afterpotentials on the recovery cycle *J. Neurophysiol.* **87** 995–1006
- McNeal D R 1976 Analysis of a model for excitation of myelinated nerve *IEEE Trans. Biomed. Eng.* **BME-23** 329–37
- Meffin H, Tahayori B, Grayden D B and Burkitt A N 2012 Modeling extracellular electrical stimulation: I. Derivation and interpretation of neurite equations *J. Neural Eng.* **9** 065005
- Meng K, Fellner A, Rattay F, Ghezzi D, Meffin H, Ibbotson M R and Kamenova T 2018 Upper stimulation threshold for retinal ganglion cell activation *J. Neural Eng.* **15** 046012
- Minassian K, Persy I, Rattay F, Pinter M M, Kern H and Dimitrijevic M R 2007 Human lumbar cord circuitries can be activated by extrinsic tonic input to generate locomotor-like activity *Hum. Mov. Sci.* **26** 275–95
- Naples G G, Mortimer J T, Scheiner A and Sweeney J D 1988 A spiral nerve cuff electrode for peripheral nerve stimulation *IEEE Trans. Biomed. Eng.* **35** 905–16
- Potrusil T, Heshmat A, Sajedi S, Wenger C, Chacko L J, Glueckert R, Schrott-Fischer A and Rattay F 2020 Finite element analysis and three-dimensional reconstruction of tonotopically aligned human auditory fiber pathways: a computational environment for modeling electrical stimulation by a cochlear implant based on micro-CT *Hear. Res.* **393** 108001
- Ranck J B 1975 Which elements are excited in electrical stimulation of mammalian central nervous system: a review *Brain Res.* **98** 417–40
- Rattay F 1990 *Electrical Nerve Stimulation* (Wien: Springer)
- Rattay F 1999 The basic mechanism for the electrical stimulation of the nervous system *Neuroscience* **89** 335–46
- Rattay F 2014 On the upper threshold phenomenon of extracellular neural stimulation *J. Neurophysiol.* **112** 2664–5
- Rattay F and Aberham M 1993 Modeling axon membranes for functional electrical stimulation *IEEE Trans. Biomed. Eng.* **40** 1201–9
- Rattay F and Resatz S 2004 Effective electrode configuration for selective stimulation with inner eye prostheses *IEEE Trans. Biomed. Eng.* **51** 1659–64
- Reichel M, Mayr W and Rattay F 1999 Computer simulation of field distribution and excitation of denervated muscle fibers caused by surface electrodes *Artif. Organs* **23** 453–6
- Romeni S, Valle G, Mazzoni A and Micera S 2020 Tutorial: a computational framework for the design and optimization of peripheral neural interfaces *Nat. Protocols* **15** 3129–53

- Stickler D I Y, Martinek J and Rattay F 2009 Modeling needle stimulation of denervated muscle fibers: voltage–distance relations and fiber polarization effects *IEEE Trans. Biomed. Eng.* **56** 2396–403
- Struijk J J, Holsheimer J, van Veen B K and Boom H B 1991 Epidural spinal cord stimulation: calculation of field potentials with special reference to dorsal column nerve fibers *IEEE Trans. Biomed. Eng.* **38** 104–10
- Tahayori B, Meffin H, Dokos S, Burkitt A N and Grayden D B 2012 Modeling extracellular electrical stimulation: II. Computational validation and numerical results *J. Neural Eng.* **9** 065006
- Tuch D S, Wedeen V J, Dale A M, George J S and Belliveau J W 2001 Conductivity tensor mapping of the human brain using diffusion tensor MRI *Proc. Natl Acad. Sci.* **98** 11697–701
- Verrills P, Sinclair C and Barnard A 2016 A review of spinal cord stimulation systems for chronic pain *J. Pain Res.* **9** 481
- Werginz P, Benav H, Zrenner E and Rattay F 2015 Modeling the response of ON and OFF retinal bipolar cells during electric stimulation *Vision Res.* **111** 170–81
- Werginz P, Raghuram V and Fried S I 2020 The relationship between morphological properties and thresholds to extracellular electric stimulation in α RGCs *J. Neural Eng.* **17** 045015